STUDENT PROJECTS LABORATORY - 930 KINNEAR RD

STANDARD OPERATING PROCEDURE (SOP) | ASSET: WLD-04 | TAG: PLASMA-CNC-01

CNC Plasma Table & Hand Plasma Station

Facility Zone: Zone 3: Welding & Thermal Processing (WLD) | Supervisor: Max Wright (Sr. Instructional Lab Supervisor)

1. REQUIRED PERSONAL PROTECTIVE EQUIPMENT (PPE)

1. Shade 5-8 Plasma Face Shield / Cutting Glasses
2. Leather Welding Gloves & FR Jacket
3. Hearing Protection
4. Fume / Water Table Downdraft Extraction Active

2. PRIMARY OPERATIONAL HAZARDS

1. High-voltage plasma arc (up to 300V DC open-circuit) and molten dross spray
2. Automated X/Y CNC gantry pinch/crush hazard
3. Hydrogen gas pocket hazard under submerged aluminum on water table

3. BEFORE OPERATION CHECKLIST

1. Verify water table water level covers slats and check slat condition.
2. Inspect torch consumables (swirl ring, electrode, nozzle, retaining cap, shield).
3. Verify dry air supply pressure (85-100 PSI) and empty moisture trap.

4. DURING OPERATION SAFETY RULES

1. Keep hands and tools completely clear of X/Y gantry travel during execution.
2. Monitor Torch Height Control (THC) and stand near E-Stop.

5. AFTER USE & SHOWROOM CLEANUP

1. Power down CNC controller and plasma supply, close air supply valve, and remove cut parts/slag with tongs.

EMERGENCY SHUTDOWN & INCIDENT PROTOCOL:

Hit red E-Stop on CNC controller or turn off plasma power supply disconnect immediately.